Evaluating the Effect of Stellar Population Models on Sodium Absorption Inflow Measurements in Elliptical Galaxies

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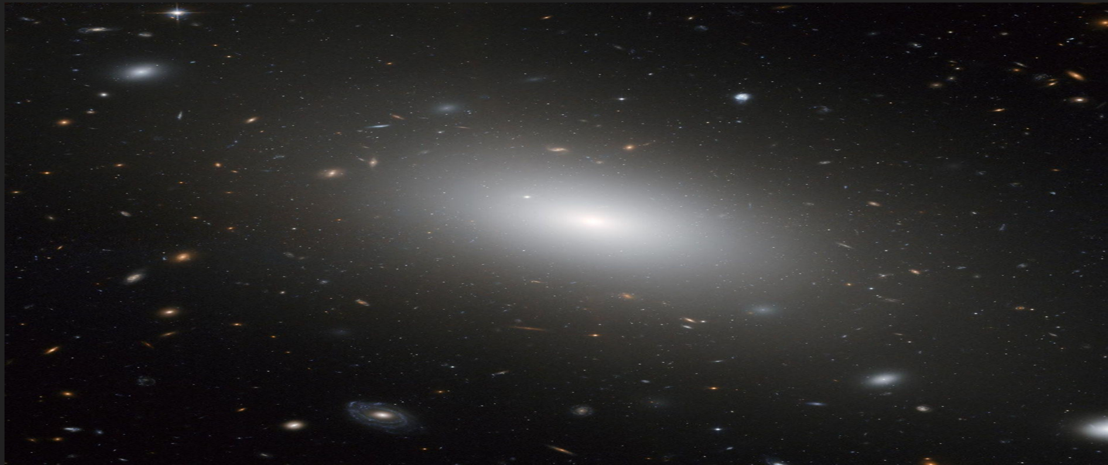
Elliptical Galaxies

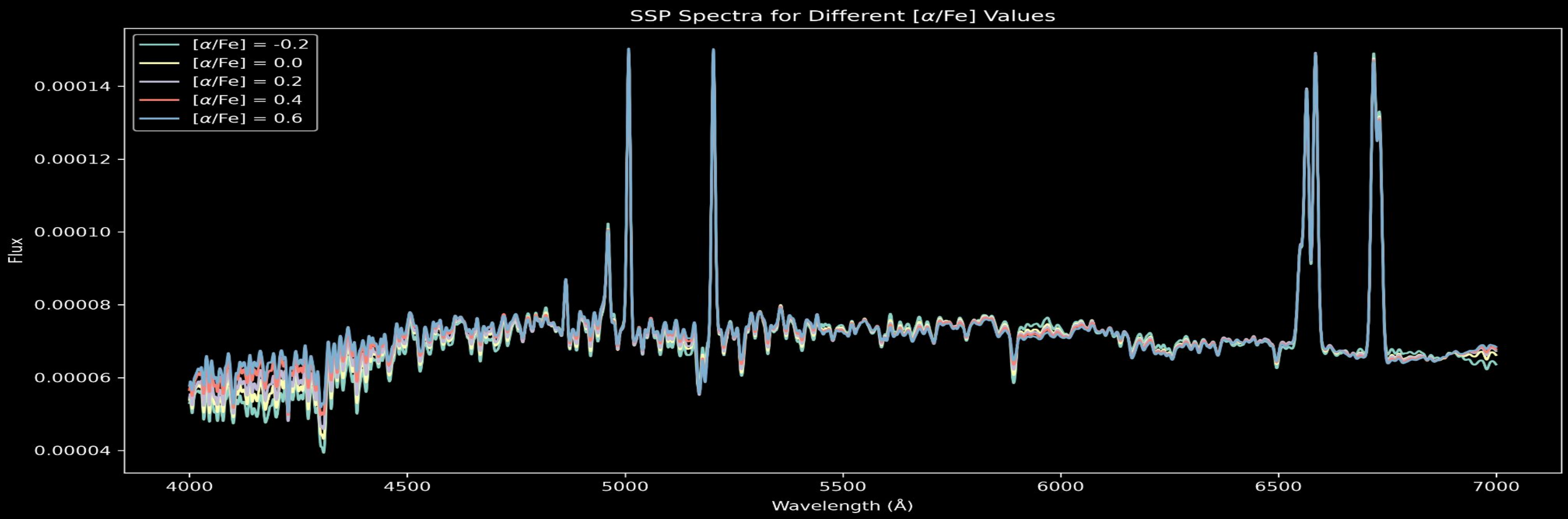
- Smooth, rounded / elliptical shape
- Home of old stars
- Stellar orbits (stars move in chaotic direction instead of an orderly flat rotation)
- Generally contain less cold gas and have less star formation than spiral galaxies



Credit: NASA/ESA, Hubble Space Telescope - NGC 1132

Alpha-to-Iron Abundance [α/Fe]

- [α/Fe] compares abundance of alpha elements to iron
- Alpha elements include oxygen, neon, magnesium, silicon, sulfur, argon, calcium, & titanium
- Different [α/Fe] change the strengths of absorption features
- Elliptical galaxies are known to have an enhanced [α/Fe] ratios (NOT ALL, but some) (Thomas et al., 1999)

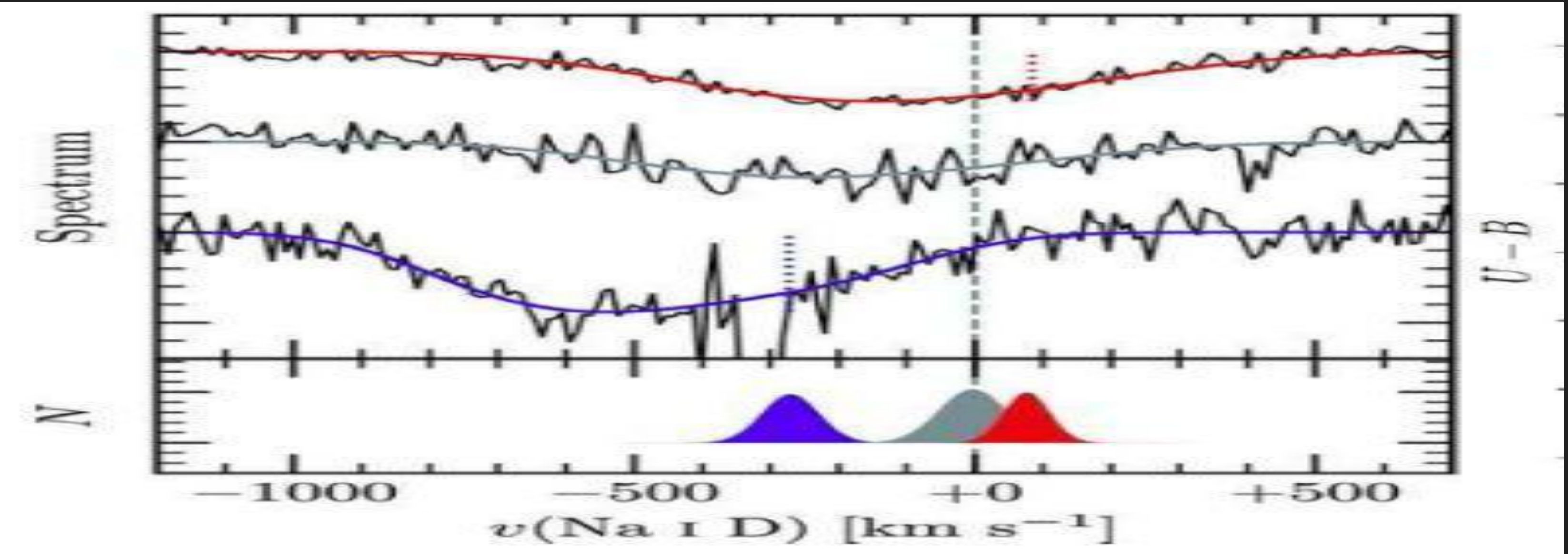


Recent Work Suggests Many Ellipticals Have Inflow

- Weng et al. 2026 **Weng et al. — main DESI/inflow paper**
  Weng, S., Saintonge, A., Pieri, M. M., et al. (2026). *Peering down the barrel with DESI DR2: 10,000+ inflows at z<0.6 reveal how galaxies accrete cold gas.* arXiv:2605.02999.
- Identified a large sample of galaxies showing cool gas inflow (elliptical galaxies included)
- Elliptical galaxies generally expected to contain relatively little cold gas
- ***Goal: determine whether stellar-population differences can affect how reliably Na I D inflow is measured***

How Inflow is Detected

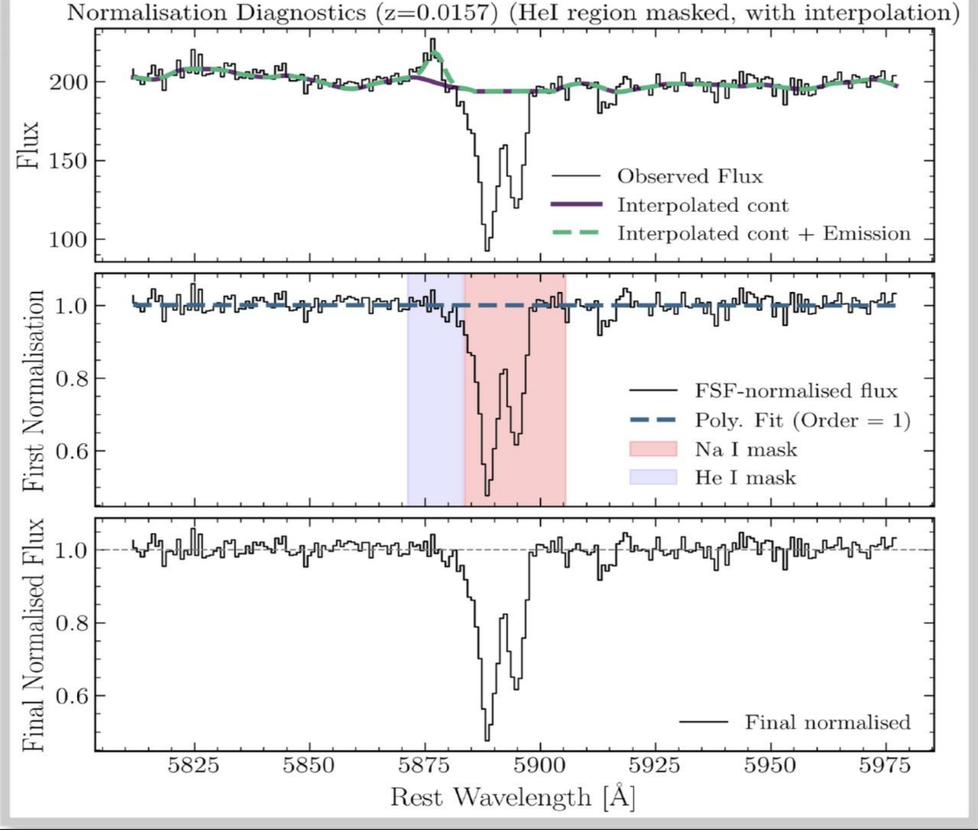
- Na I D is a strong absorption doublet near 5890 Å
- Gas absorbs light at wavelengths determined by its motion
- Compare the Na I D velocity to the galaxy’s systemic velocity
- Redshifted Na I D absorption can indicate gas moving inward



Sato et al. (2009)

Separating Stellar Na I D Absorption

- Na I D absorption can come from stars & interstellar gas
- Compare stellar Na I D across [α/Fe] values
- **Stellar absorption MUST be modeled before identifying gas-related absorption**
- ***<u>Goal: identify excess Na I D absorption that may trace inflow</u>***



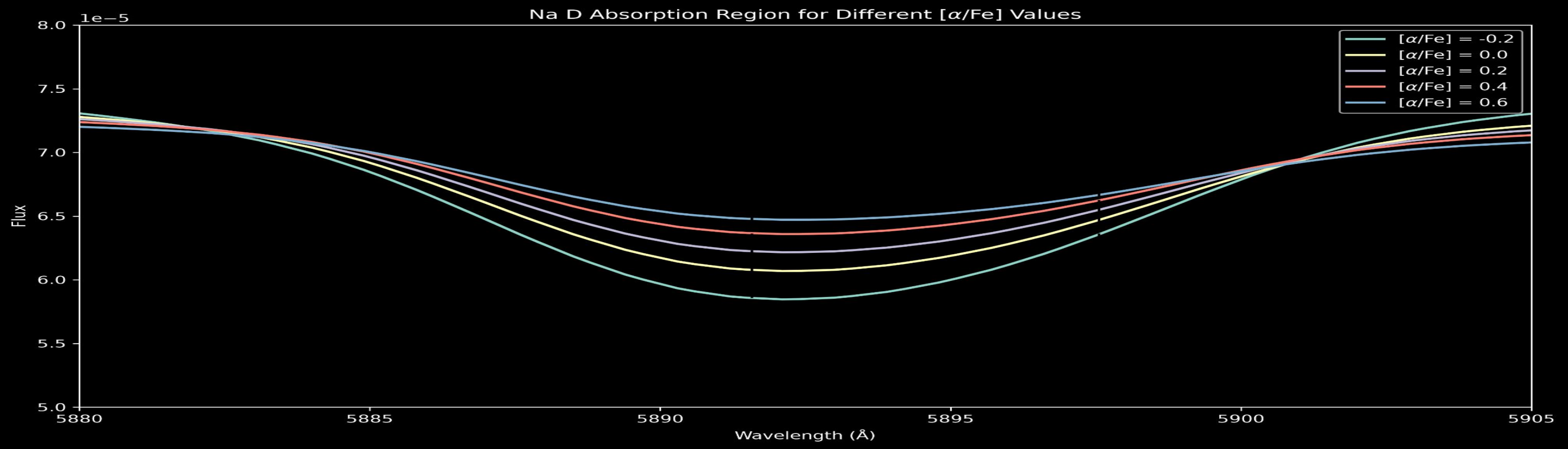
Weng et al. (2026), Fig. 1

ALfα Stellar Population Models

- Python-based stellar population modeling software
- Models absorption-line spectra of old stellar populations
- Allows individual elemental abundances to be varied
- Used to generate/model spectra with different [α/Fe] values
- Data is the spectrum of an elliptical galaxy that is distributed with the Alfα package

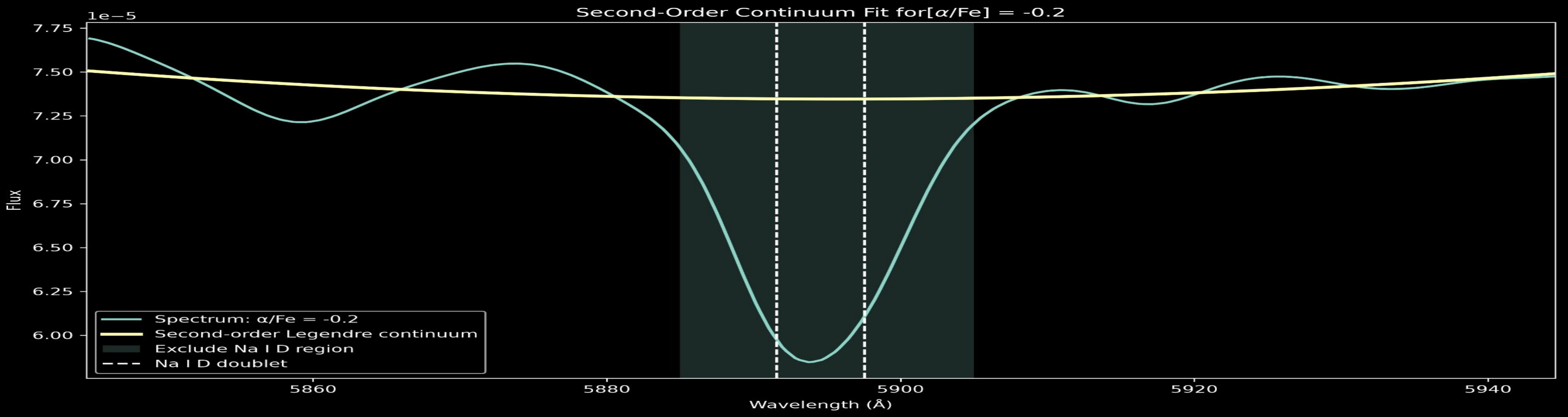
Model Stellar Spectra

- Focused on the Na I D region near 5890 Å
- Used the same wavelength on each [α/Fe] model
- Isolated the local wavelength region around the doublet
- ***Zooming in reveals abundance-dependent differences that are difficult to see in the full spectrum***

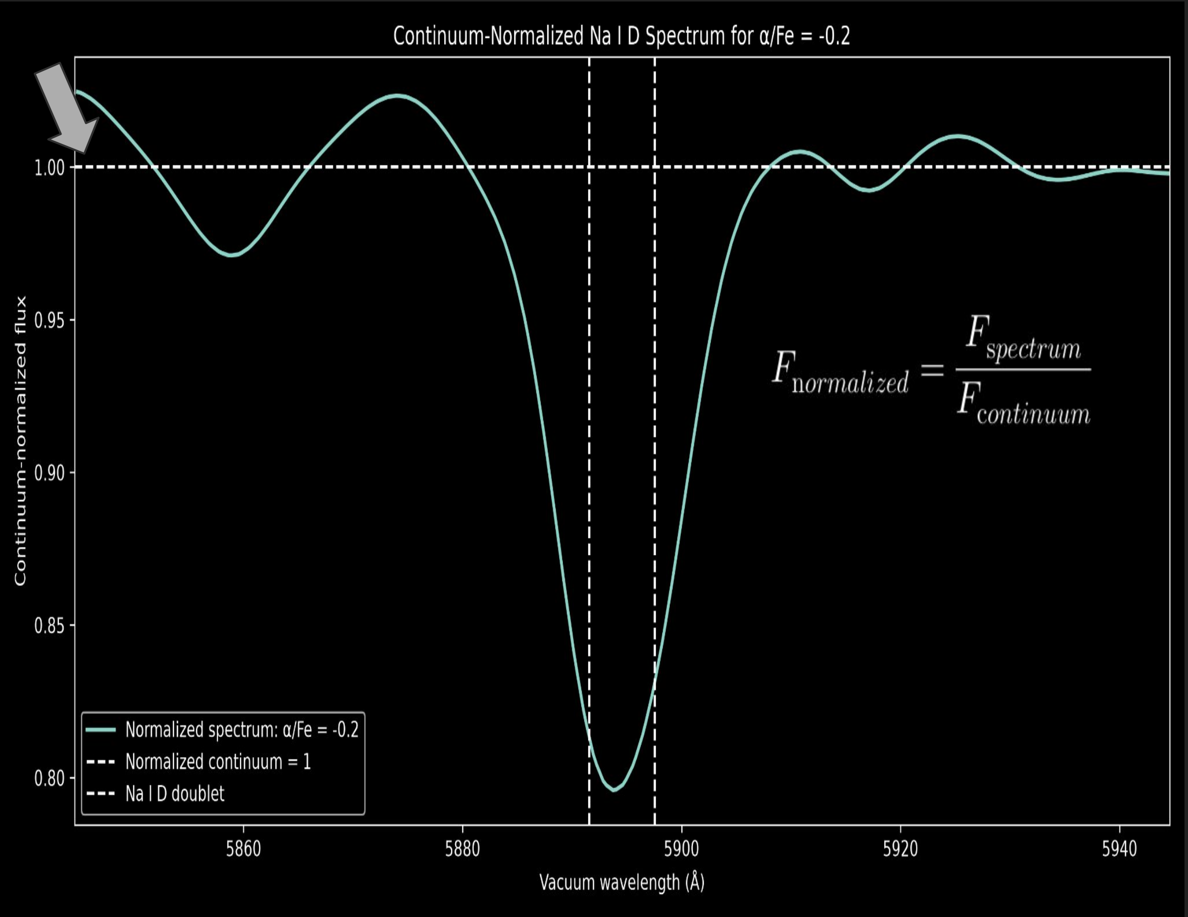


Fitting the Local Continuum Around Na I D

- Selected a local wavelength window around Na I D
- Excluded the Na I D absorption region from the fit
- Second-order Legendre polynomial for the continuum
    - ***estimates the local baseline underneath the absorption feature***
- Divided the spectrum by the fitted continuum to normalize it

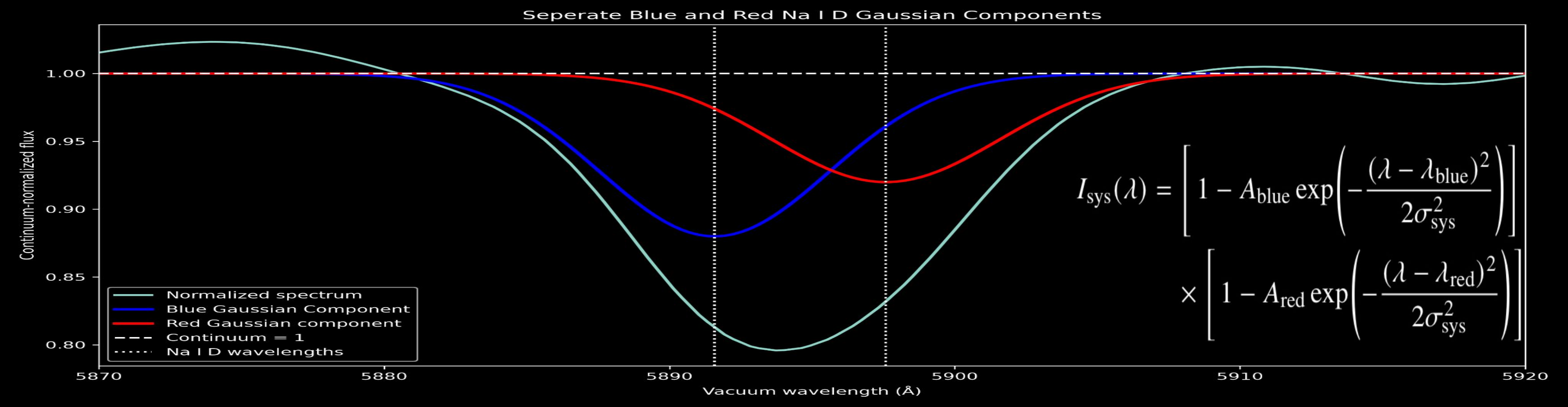


Continuum-Normalized Na I D Spectrum



- Continuum is normalized to ~ 1, provides a common reference level
- Na I D absorption depth easier to compare between models
- Normalization removes differences in the overall continuum shape & isolates the Na I D feature
- ***Allows direct comparison of absorption depths***

Modeling Na I D with the Double Gaussian

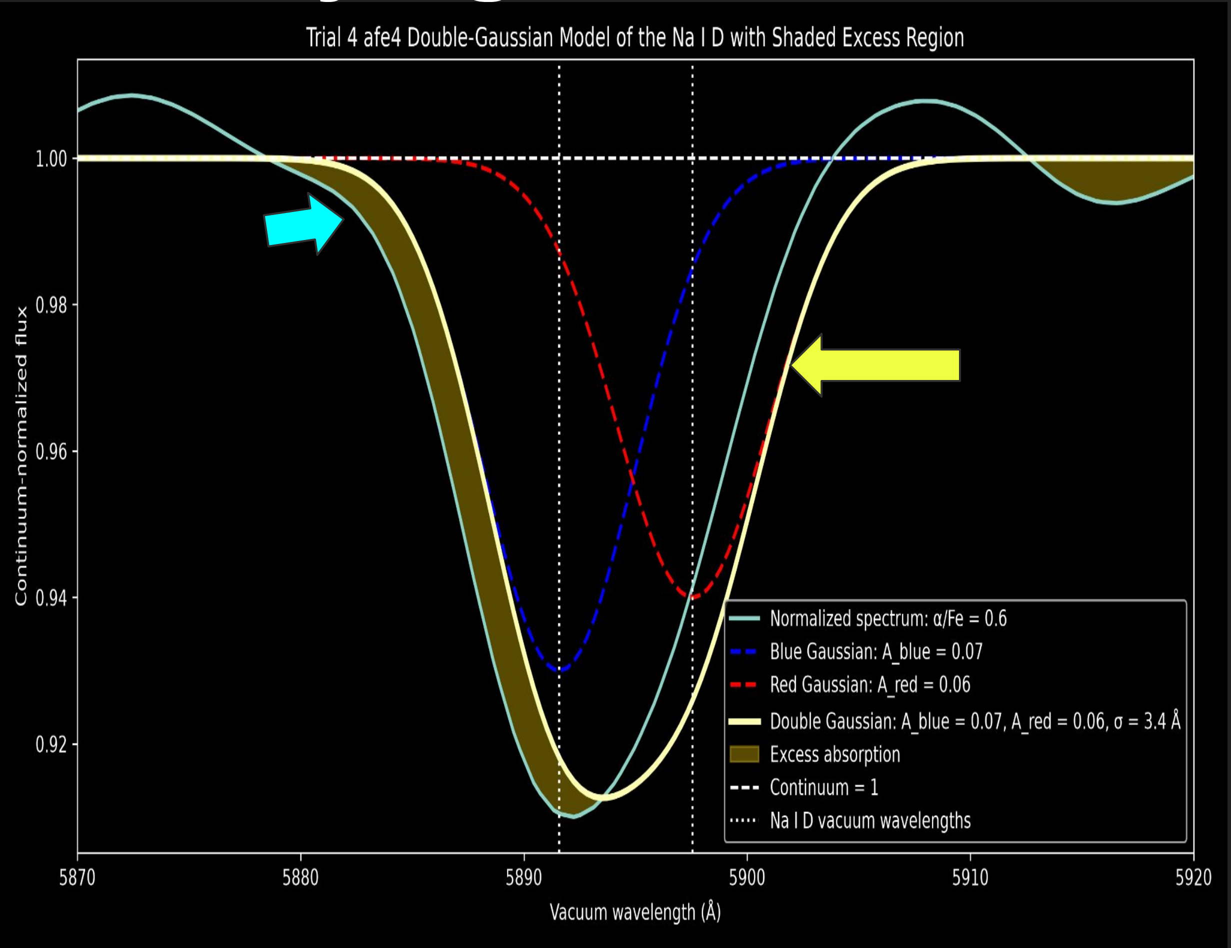


$$I_{\mathrm{sys}}(\lambda) = \left[1 - A_{\mathrm{blue}} \exp\left(-\frac{(\lambda - \lambda_{\mathrm{blue}})^2}{2\sigma_{\mathrm{sys}}^2}\right)\right] \times \left[1 - A_{\mathrm{red}} \exp\left(-\frac{(\lambda - \lambda_{\mathrm{red}})^2}{2\sigma_{\mathrm{sys}}^2}\right)\right]$$

Adapted from Weng et al. (2026) Eq. C1

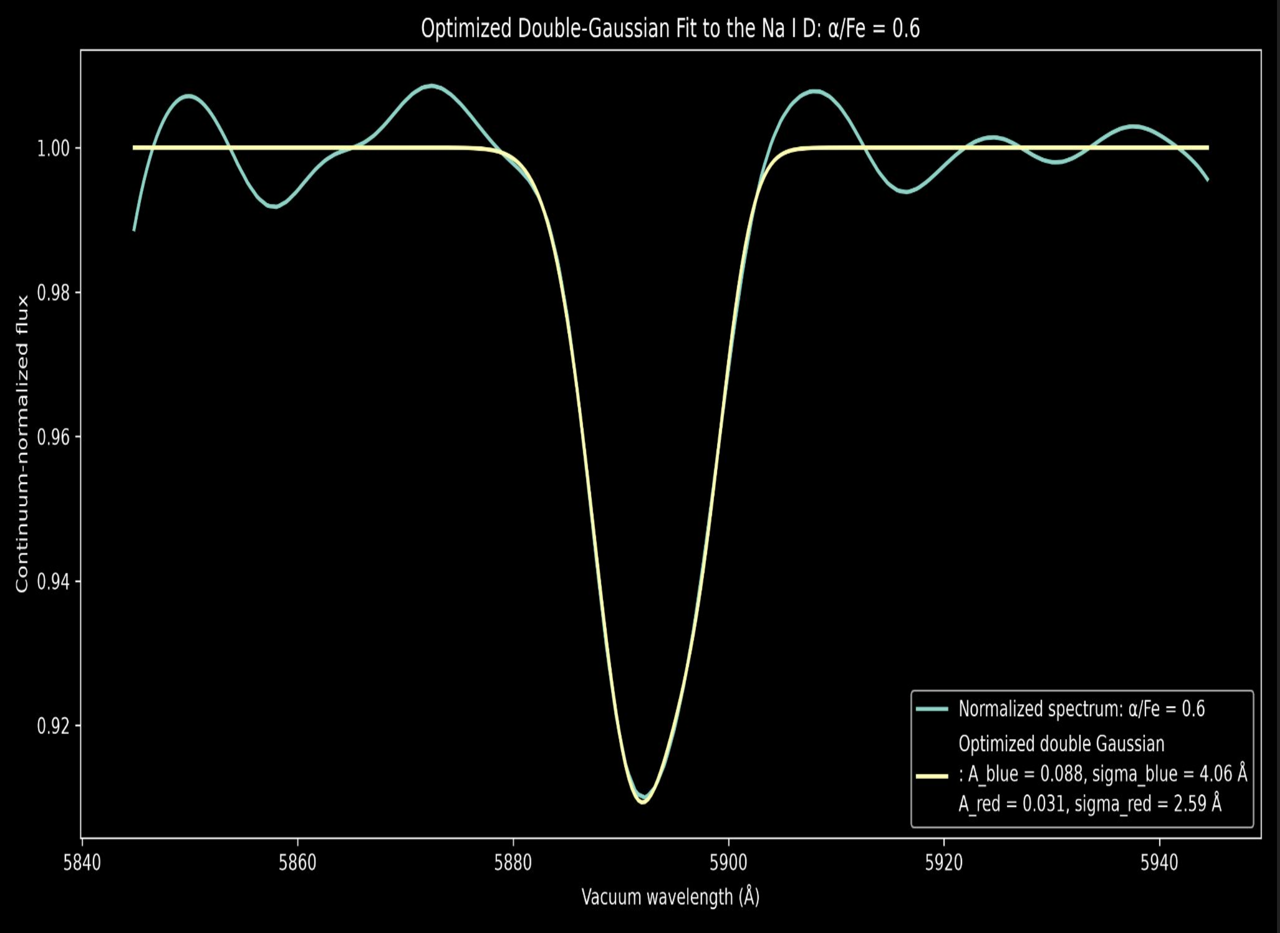
- Doublet with two nearby absorption lines
- Each line modeled with a Gaussian
- Gaussian amplitude controls absorption depth
- Gaussian width (σ) controls how broad the line is
- The two components were combined to model the full Na I D feature
- Fitted amplitudes & widths provide quantitative measurements of the stellar Na I D Profile
- ***<u>Quantitatively describes the stellar Na I D profile</u>***

Identifying Excess Na I D Absorption



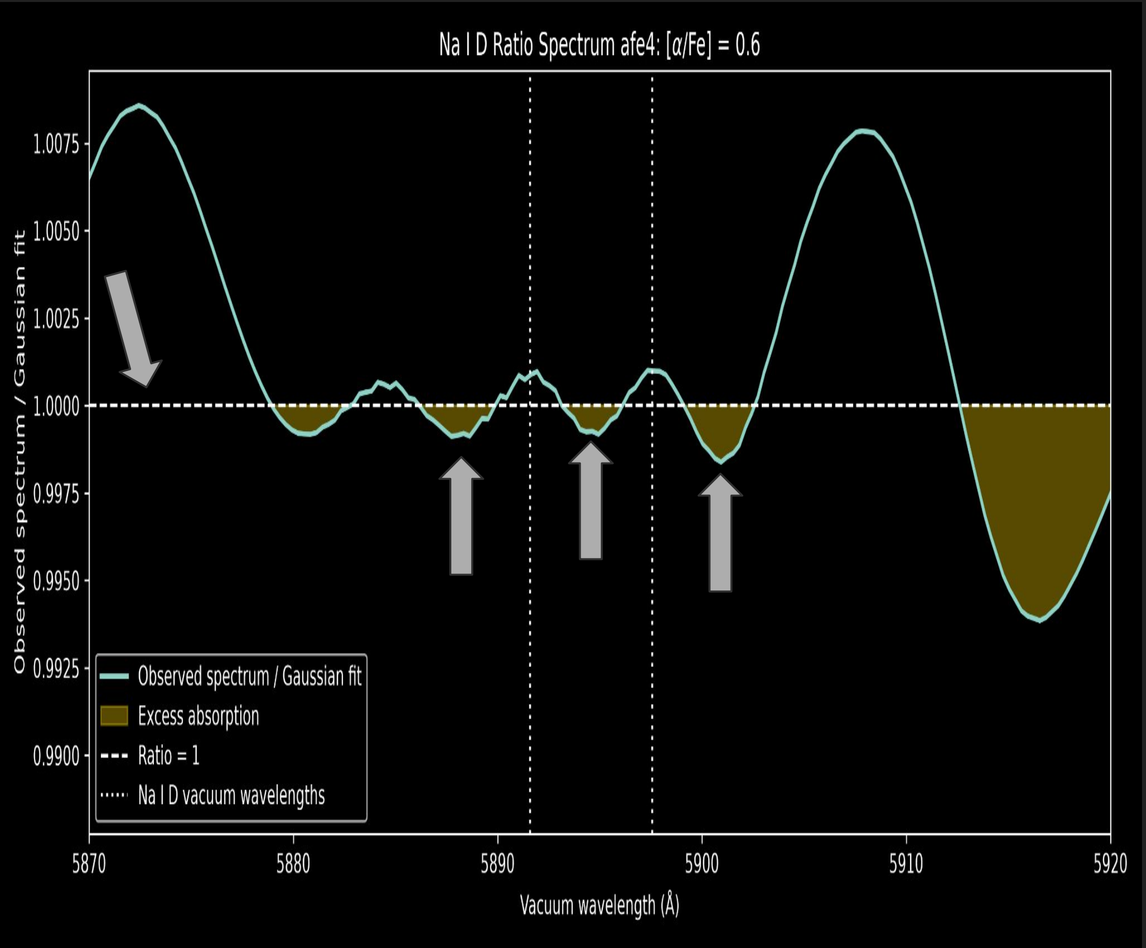
- Manually adjusted amplitudes & widths to understand how each parameter changes the fit
- Compared the manual double-Gaussian model with the normalized spectrum
- ***Shaded regions show where the spectrum has more absorption than the Gaussian model***
- Helped establish reasonable starting parameters for the optimized fit

Optimized Double-Gaussian Fit



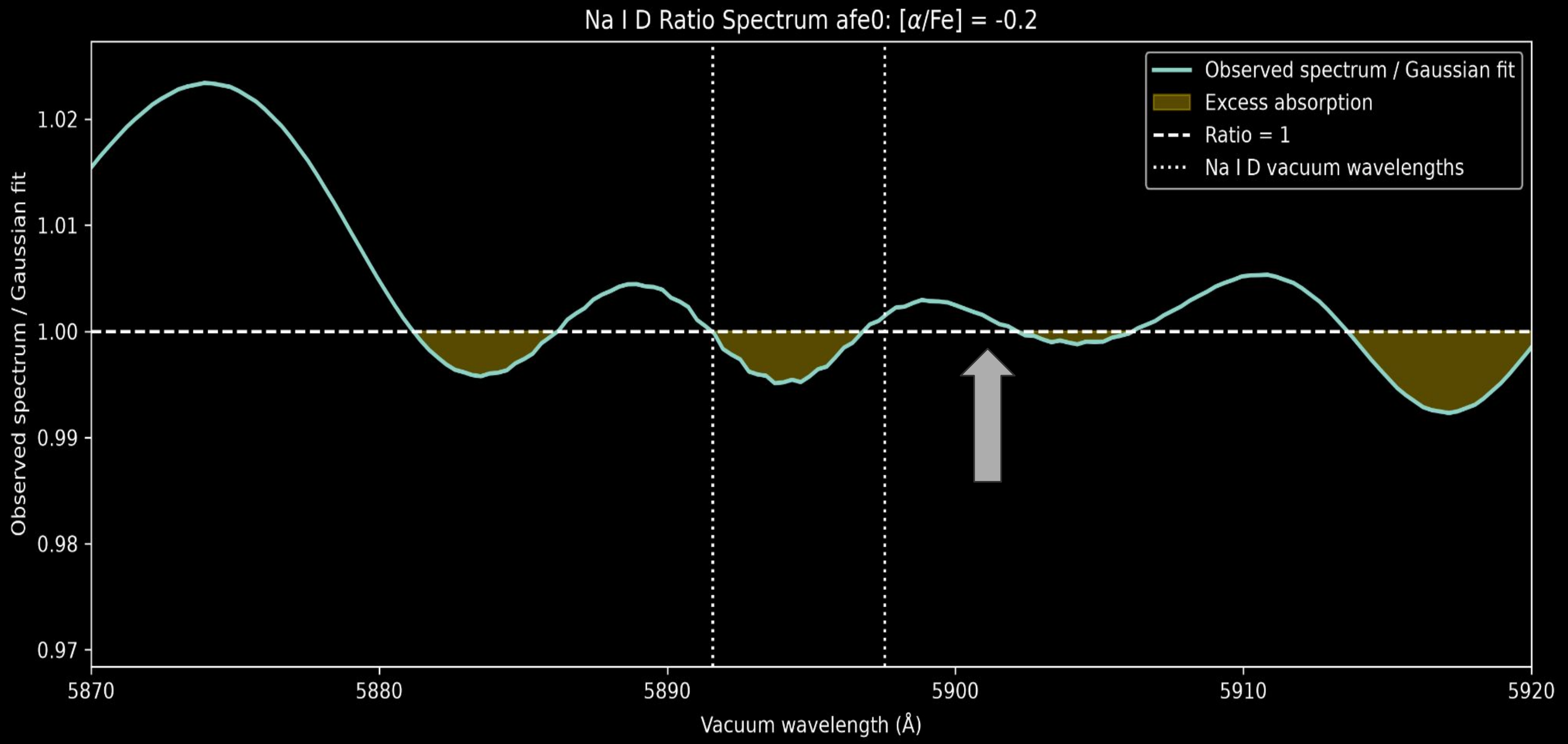
- Curve_fit optimized both Na I D absorption components simultaneously
- ***<u>Returned best-fit amplitudes</u>***
- ***<u>Optimized model closely reproduces the main stellar Na I D absorption feature</u>***
- ***<u>Fitted parameters allows the abundance models to be compared quantitatively</u>***

Residual Absorption After Gaussian Fitting

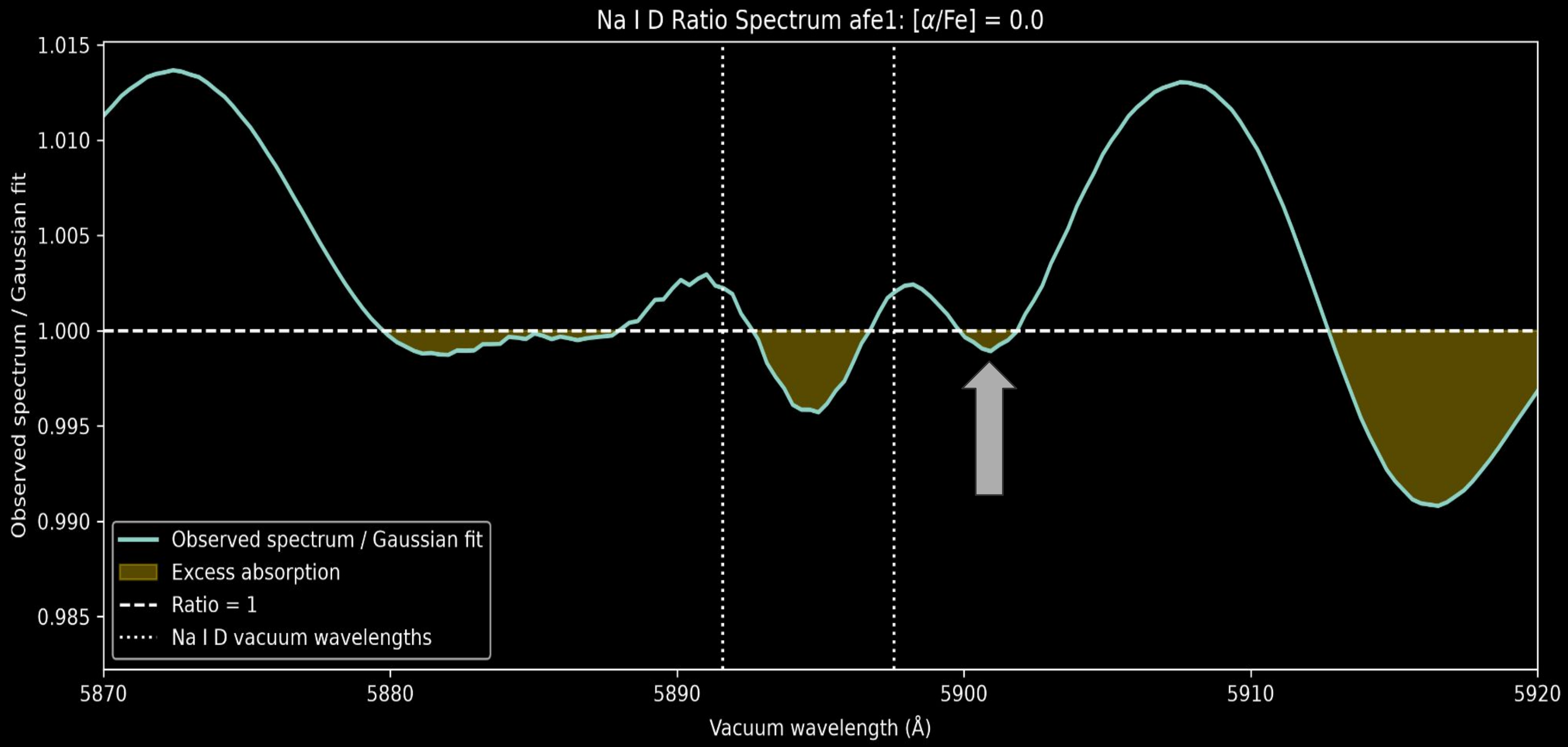


- Ratio = normalized stellar spectrum / optimized Gaussian fit
- Ratio = 1: spectrum & Gaussian model match
- Ratio < 1: actual spectrum has more absorption than Gaussian
- Ratio > 1: actual spectrum has less absorption than Gaussian
- ***Residual structure reveals where the simple double-Gaussian model does not fully reproduce the stellar spectrum***

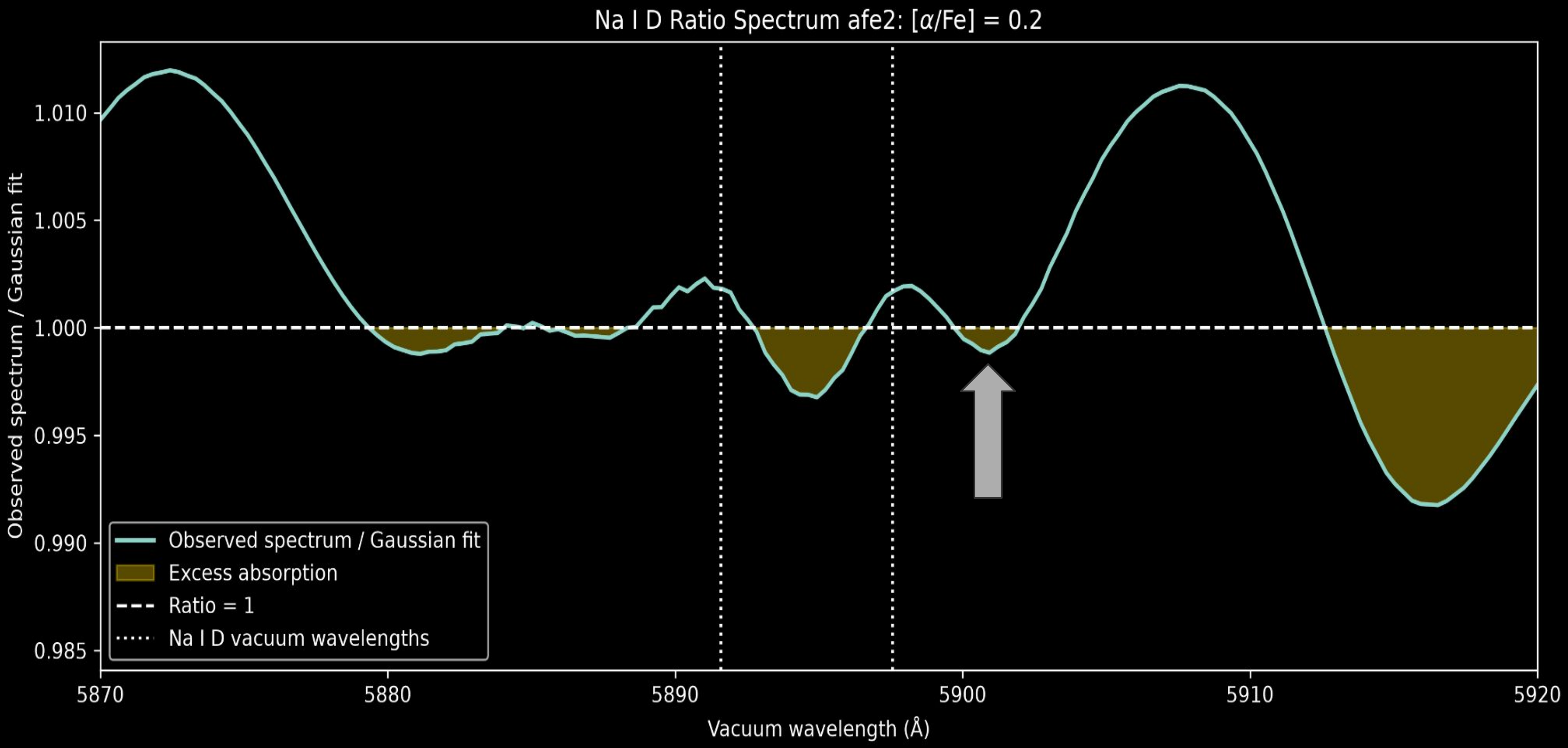
Residual Absorption After Gaussian Fitting



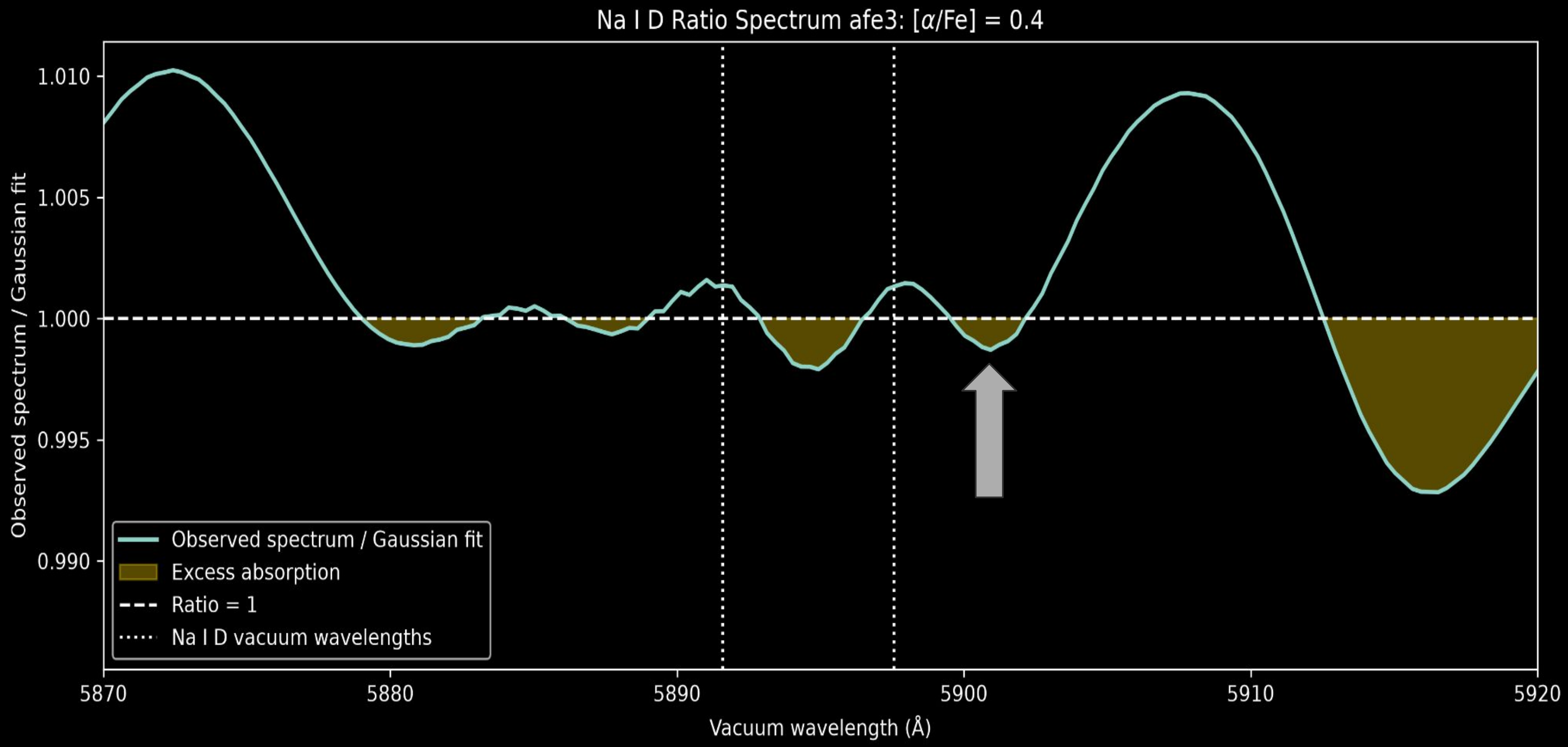
Residual Absorption After Gaussian Fitting



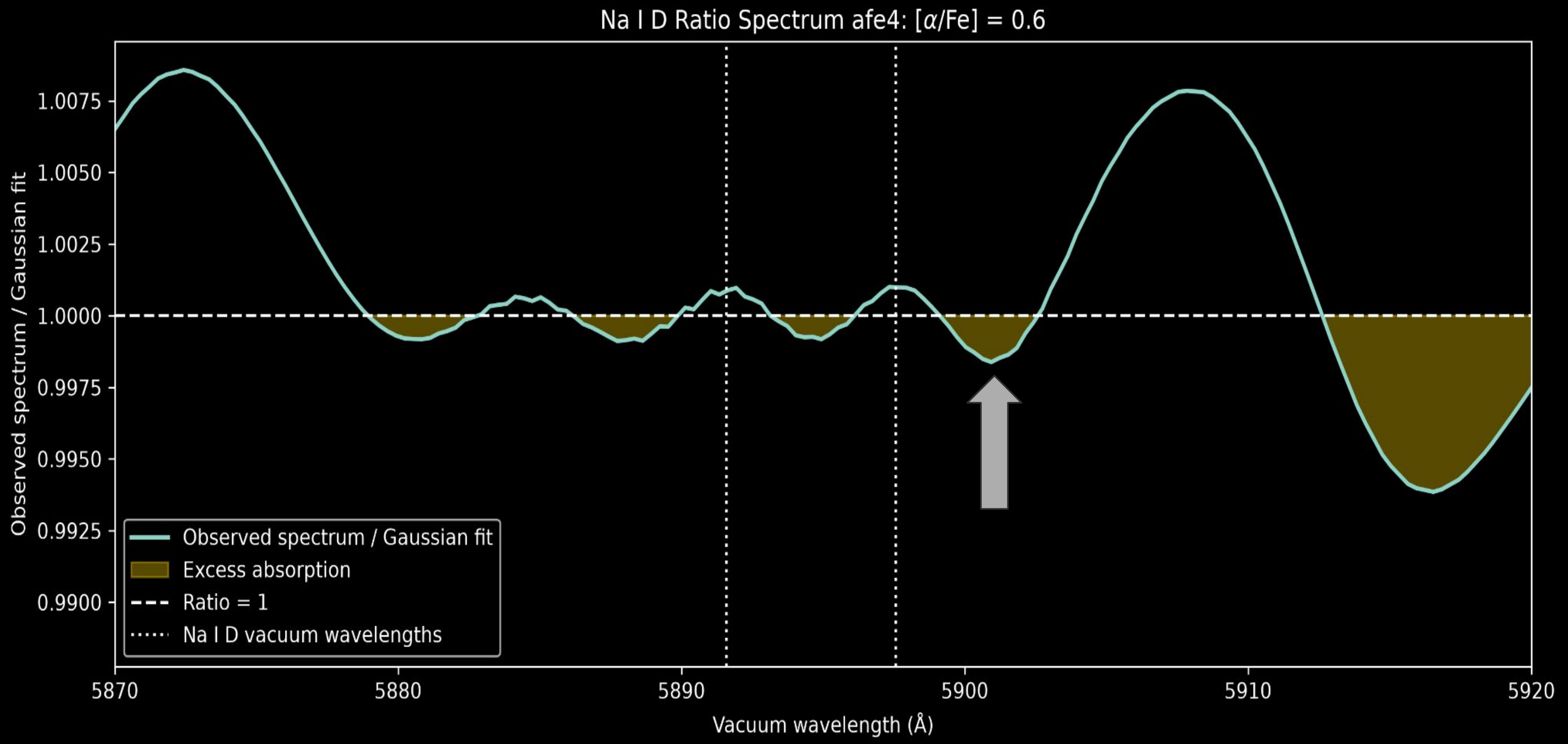
Residual Absorption After Gaussian Fitting



Residual Absorption After Gaussian Fitting



Residual Absorption After Gaussian Fitting



What the Fits Show

- ***Double-Gaussian model reproduces the dominant stellar Na I D feature***
- ***Fitted Na I D components vary between [α/Fe] abundance models***
- ***Assumed stellar population can affect how excess Na I D absorption is interpreted***
- ***Accurate stellar modeling is necessary before identifying excess gas absorption***

Conclusion

- ***Stellar Na I D changes [α/Fe], so stellar-population modeling matters when searching for excess Na I D from inflowing gas***
- Na I D contains a significant stellar absorption component
- Double-Gaussian fitting can characterize the stellar Na I D feature
- Modeling the stellar contribution helps distinguish excess absorption from gas
- ***Seeing tentative evidence that enhanced [α/Fe] ratios give rise to enhanced absorption on the red side of the Na I D profile***
- ***Future Work: Is the enhanced absorption enough to cause anyone to detect a false inflow?***

Acknowledgements

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Citations

Weng et al. — main DESI/inflow paper
Weng, S., Saintonge, A., Pieri, M. M., et al. (2026). *Peering down the barrel with DESI DR2: 10,000+ inflows at z<0.6 reveal how galaxies accrete cold gas.* arXiv:2605.02999.

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Questions?

Credit: NASA/ESA/CSA/J. DePasquale (STScI)